

CO3 – AT3 MCQ Test

Database Management-CSA0509

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Question 1

Which of the following is true about 1NF?

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic
- (d) Eliminates transitive dependencies

Correct Answer: (c) All attributes must be atomic

Question 2

A relation is in 2NF if it is in 1NF and:

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Correct Answer: (b) Has no partial dependencies on primary key

Question 3

Which normal form deals with multi-valued dependencies?

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

Correct Answer: (d) 4NF

Question 4

BCNF is a stricter version of:

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 4NF

Correct Answer: (c) 3NF

Question 5

Functional dependency $X \rightarrow Y$ means:

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

Correct Answer: (b) X determines Y

Question 6

Which of the following anomaly is NOT associated with unnormalized relations?

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly

Correct Answer: (d) Retrieval anomaly

Question 7

A superkey is:

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Correct Answer: (b) Any set of attributes that uniquely identifies tuples

Question 8

If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A
- (d) Independently determined

Correct Answer: (c) Transitively dependent on A

Question 9

5NF is also known as:

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form
- (d) Element Normal Form

Correct Answer: (c) Project-Join Normal Form

Question 10

Lossless-join decomposition ensures:

- (a) No data is lost during decomposition
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

Correct Answer: (a) No data is lost during decomposition

Answer Key

1. (c) All attributes must be atomic
2. (b) Has no partial dependencies on primary key
3. (d) 4NF
4. (c) 3NF
5. (b) X determines Y
6. (d) Retrieval anomaly
7. (b) Any set of attributes that uniquely identifies tuples
8. (c) Transitively dependent on A
9. (c) Project-Join Normal Form
10. (a) No data is lost during decomposition